2. Arithmetic operations

Summary

No.	Tags	Code
1	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (add, addition)), (math, mathematics)	`x + y`
2	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (subtract)), (math, mathematics)	`x - y`
3	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (multiplication, product)), (math, mathematics)	`x * y`
4	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (fraction, division)), (math, mathematics)	`x / y`
5	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (add, addition)), (vs, memory usage), (math, mathematics)	`x.add_(y)`
6	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (subtract)), (vs, memory usage), (math, mathematics)	`x.sub_(y)`
7	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (multiplication, product)), (vs, memory usage), (math, mathematics)	`x.mul_(y)`
8	Python, PyTorch, (tutorial, how-to), (arithmetic, operations, vector arithmetic, (fraction, division)), (vs, memory usage), (math, mathematics)	`x.div_(y)`

Examples

> Element-wise arithmetic operations on tensors are performed as follows.
>
> All of the operations create new tensors.
>
> In-place operations (that modify the original tensor) are performed using the following methods with the _ suffix, as shown below.

In [23]:
```
import torch
```

In [24]:
```
x = torch.tensor([3, 2, 1])
y = torch.tensor([1, 2, 3])
```

Operations

In [25]:
```
x + y         # Addition
```

Out[25]:
```
tensor([4, 4, 4])
```

In [26]:
```
x - y    # Subtraction
```

Out[26]:
```
tensor([ 2,  0, -2])
```

In [27]:
```
x * y    # Multiplication
```

Out[27]:
```
tensor([3, 4, 3])
```

In [28]:
```
x / y      # Division
```

Out[28]:
```
tensor([3.0000, 1.0000, 0.3333])
```

In-place operations

In [29]:
```
x
```

Out[29]:
```
tensor([3, 2, 1])
```

In [30]:
```
# In-place operations (modifies x)
x.sub_(3)
```

Out[30]:
```
tensor([ 0, -1, -2])
```

In [31]:
```
x
```

Out[31]:
```
tensor([ 0, -1, -2])
```